Day 4: Hands-on Subnetting Practical (1 hour): Divide a network into subnets and assign IP addresses to devices.Task: Document the steps to create a subnet plan.

Tasks

1.Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

ANS:

1. Calculation

- Given Network: 192.168.10.0/24
- Required Subnets: 4
- Borrowed Bits: 2
- New CIDR: /26
- Subnet Mask: 255.255.255.192

Calculation:

Number of subnets = \(2^2 = 4\)

Each subnet has 64 total IP addresses.

2. Subnet Table

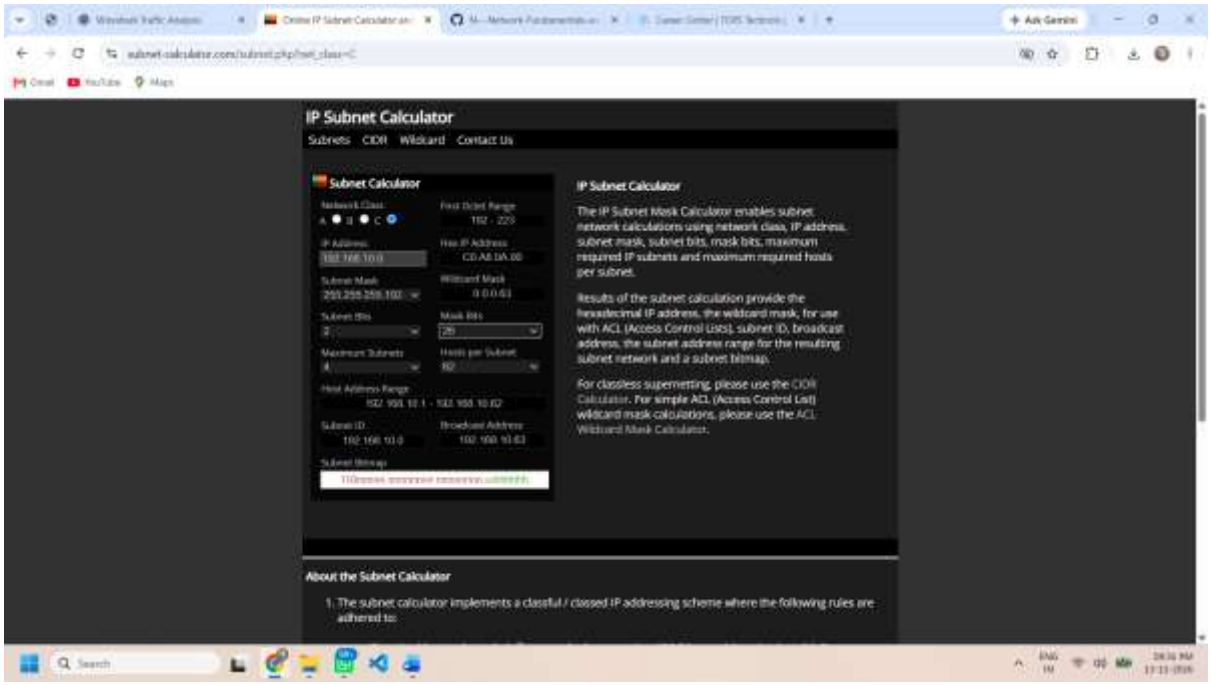
Subnet	Network Address	Subnet Mask	Broadcast Address
1	192.168.10.0	255.255.255.192	192.168.10.63
2	192.168.10.64	255.255.255.192	192.168.10.127
3	192.168.10.128	255.255.255.192	192.168.10.191
4	192.168.10.192	255.255.255.192	192.168.10.255

2.For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.

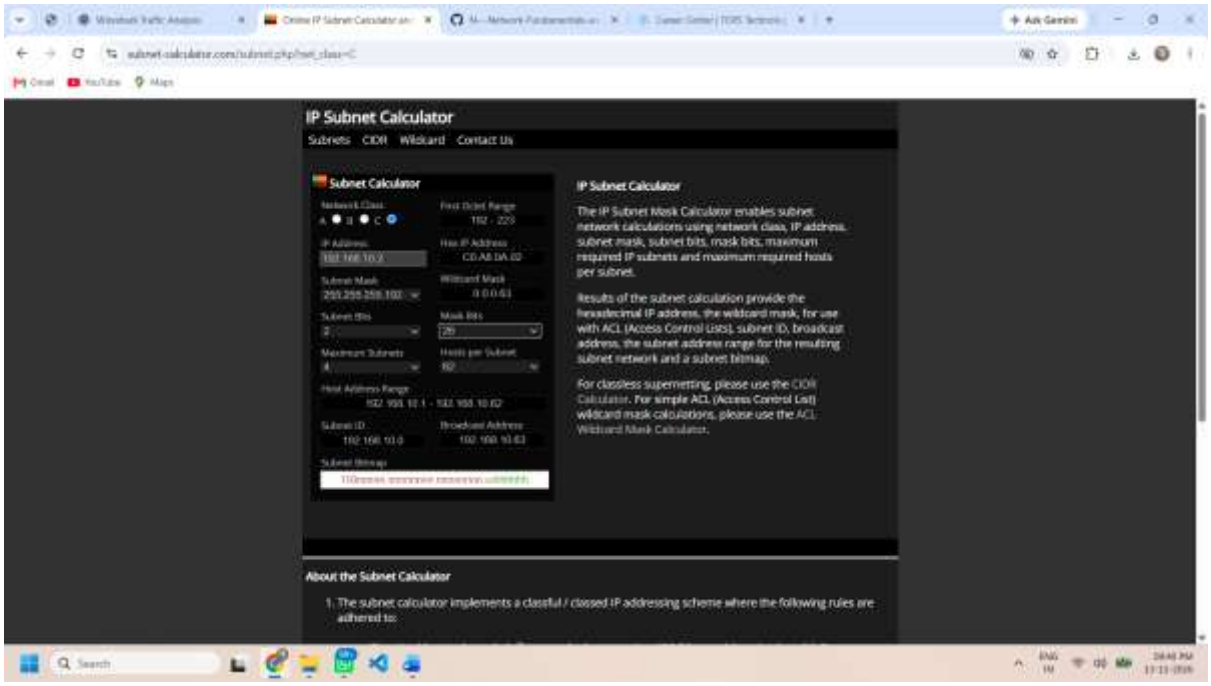
ANS:

Router

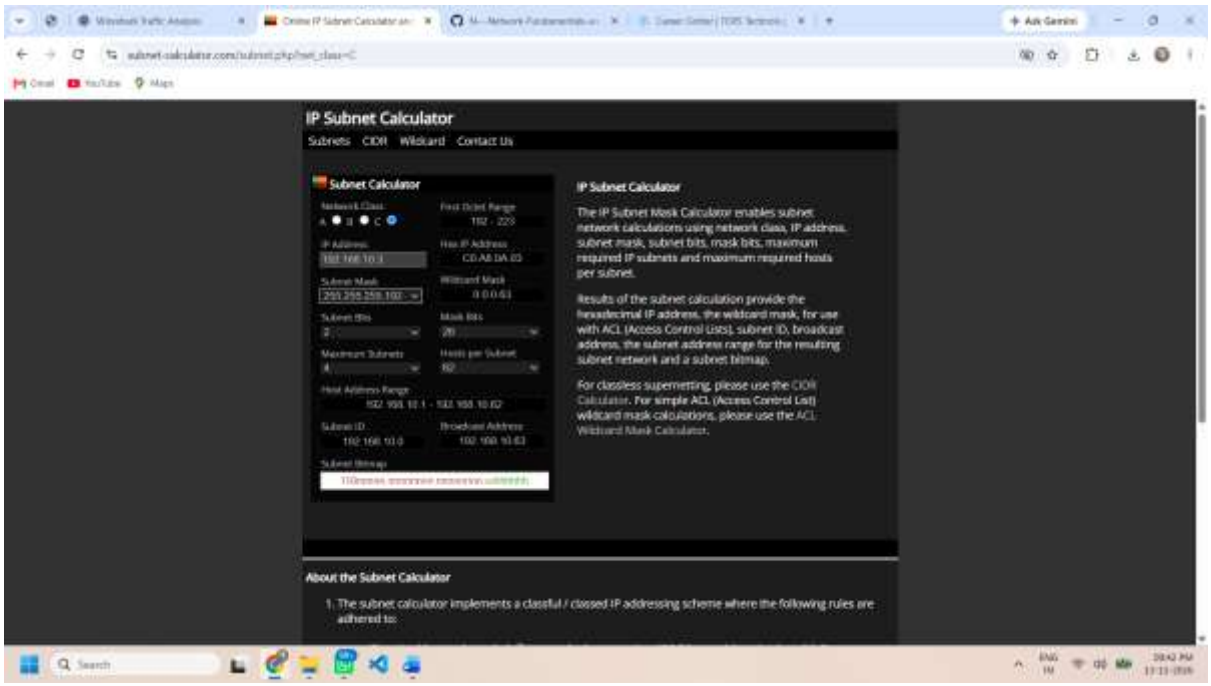


Details	**Your Screenshot**
IP Address	192.168.10.0
Subnet Mask	255.255.255.192
CIDR	/26
Subnet Bits	2
Maximum Subnets	4
Hosts per Subnet	62
Network Address	192.168.10.0
Broadcast Address	192.168.10.63
Valid Host Range	192.168.10.1 – 192.168.10.62

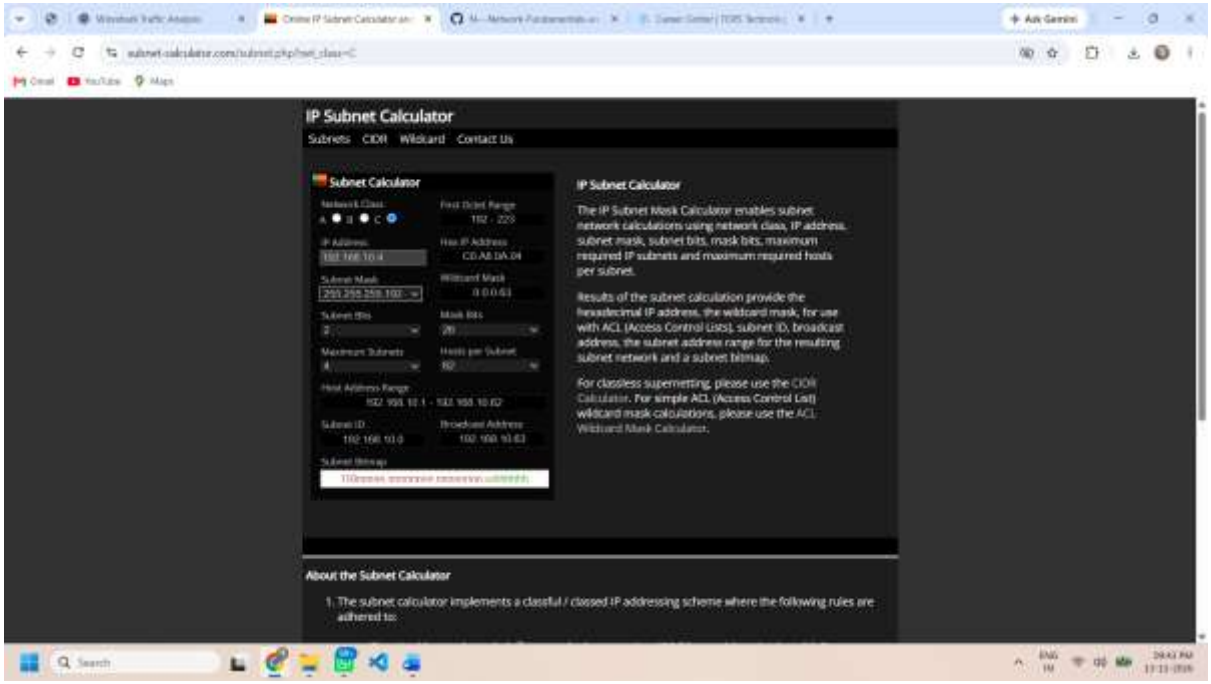
Laptop 1



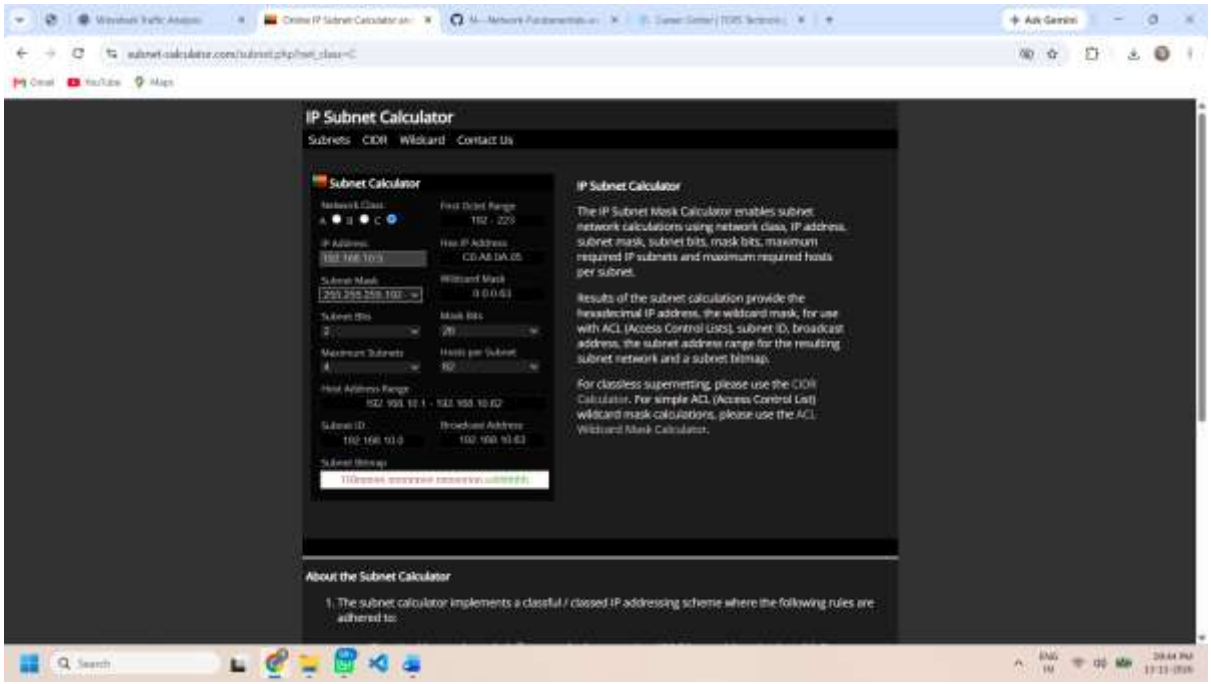
Laptop 2



Printer

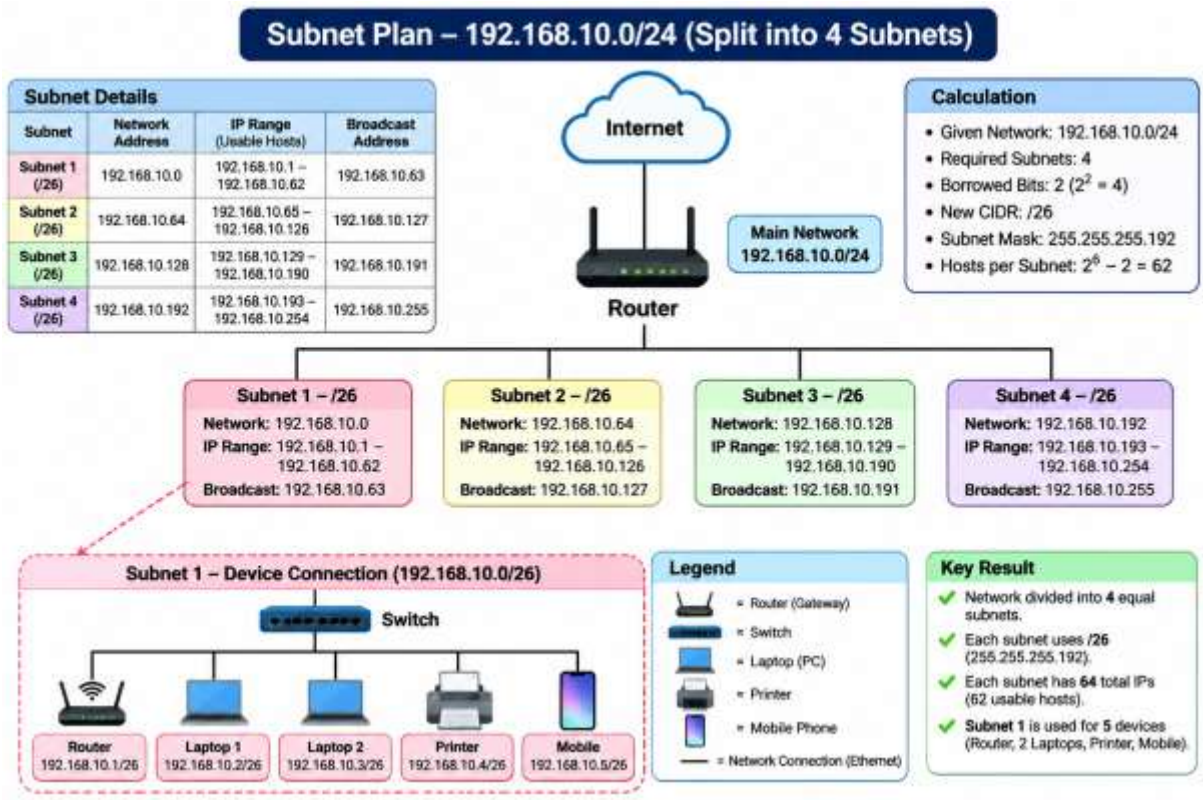


Mobile



3.Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

ANS:



4.Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

ANS:

Step 1: Given Information

- Network: 192.168.10.0/24
- Required Subnets: 4
- Devices: 5

Step 2: Calculate Subnets

Formula:

\[2^2=4 \]

We borrow 2 bits to create 4 subnets.

Step 3: Find New CIDR

\[24+2=26 \]

New CIDR = /26

Step 4: Find Subnet Mask

Subnet Mask = 255.255.255.192

Step 5: Calculate Total IP Addresses

\[2^{32-26}=64 \]

Total IP addresses = 64.

Step 6: Calculate Usable Hosts

\[64-2=62 \]

Usable IP addresses = 62.

Step 7: Divide into 4 Subnets

Subnet	Network Address	Broadcast Address
1	192.168.10.0	192.168.10.63
2	192.168.10.64	192.168.10.127
3	192.168.10.128	192.168.10.191
4	192.168.10.192	192.168.10.255

Step 8: Assign IP Addresses

Device	IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

Step 9: Default Gateway

Default Gateway for all devices = 192.168.10.1 (Router).